

Sir Syed University of Engineering & Technology
Department of Computer Science & Information Technology



HS-411 - Entrepreneurship
Lecture No. 6

Lean Startup Methodology

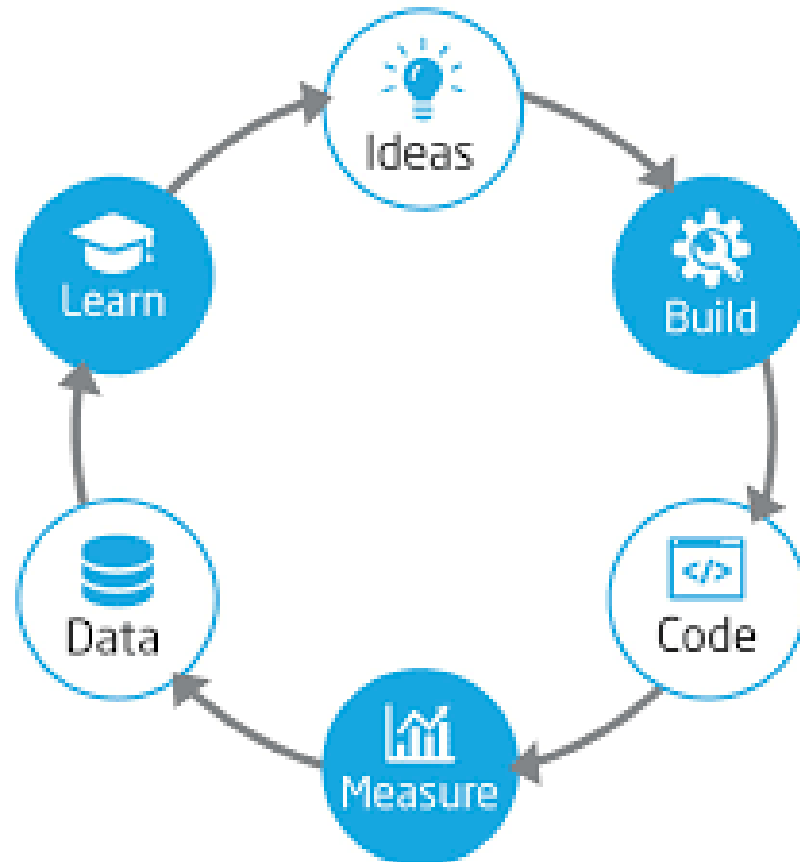
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List of Topic to be Discuss in this lecture

1. Lean Startup Principles
2. Key Principles of the Lean Startup
3. Minimum Viable Product (MVP)
4. Build-Measure-Learn Cycle
5. Product/Market Fit
6. Problem/Solution Fit

Lean Startup Methodology

The Lean Startup methodology, developed by Eric Ries, is a framework for building and managing startups based on principles of **continuous innovation**, **validated learning**, and **rapid iteration**.



Key Principles of the Lean Startup

Key principles of the Lean Startup approach along with examples:

Build-Measure-Learn: Instead of spending significant time and resources developing a product without feedback, the Lean Startup advocates for quickly building a minimum viable product (MVP), measuring its performance, and learning from user feedback to make informed decisions.

For example, Dropbox initially created a simple MVP—a video demonstrating how the product would work—before fully developing the file-sharing platform. They measured user interest through sign-ups and feedback and learned that there was significant demand for their product.

Key Principles of the Lean Startup

Validated Learning: The Lean Startup emphasizes the importance of validating assumptions and hypotheses through experiments and data rather than relying on guesswork.

For Example, before launching Airbnb, the founders tested their assumption that people would be willing to rent out their homes to strangers by creating a simple website offering air mattresses for rent during a conference. The positive response validated their idea and encouraged them to continue developing the platform.

Key Principles of the Lean Startup

Minimum Viable Product (MVP): An MVP is the simplest version of a product that allows a team to collect the maximum amount of validated learning about customers with the least effort.

For example, before creating its electric cars, Tesla Motors built the Tesla Roadster as its MVP—a high-performance electric sports car based on a pre-existing vehicle platform. The Roadster's success provided the foundation for further development and expansion into the electric vehicle market.

Key Principles of the Lean Startup

Continuous Deployment: Lean Startup encourages frequent and incremental product releases to gather feedback and iterate quickly.

For example, Facebook regularly updates its platform with new features and improvements based on user feedback and data analysis. This continuous deployment allows them to respond to user needs and market changes rapidly.

Key Principles of the Lean Startup

Pivot or Persevere: A pivot involves changing a fundamental aspect of the business strategy based on feedback and learning. Lean Startup advocates for being open to pivoting if the initial approach isn't working.

For example, Instagram started as a location-based social network called Burbn but pivoted to focus exclusively on photo sharing after noticing that users were more interested in that feature.

Key Principles of the Lean Startup

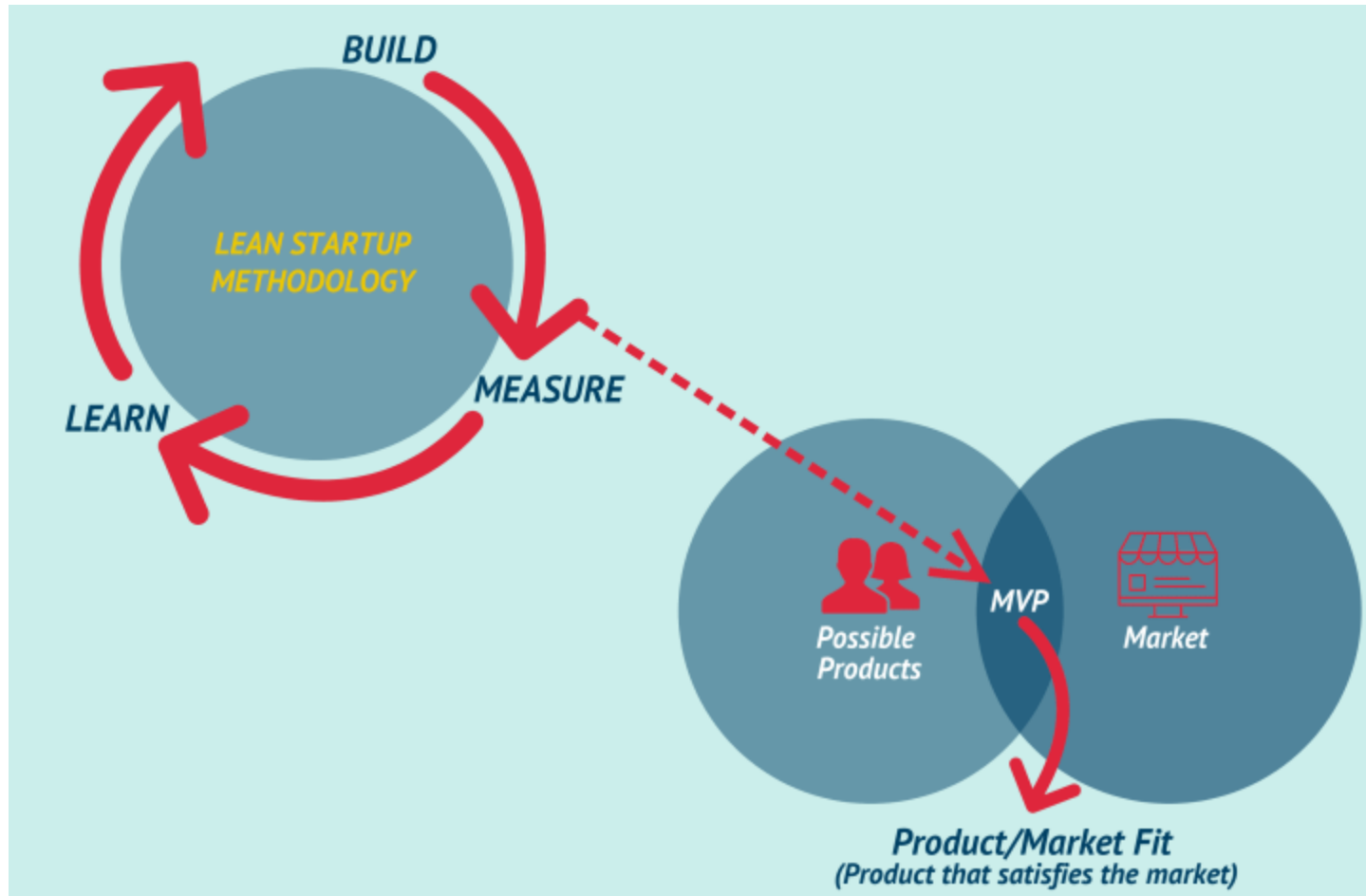
Fail Fast, Fail Cheap: Instead of viewing failure as a negative outcome, Lean Startup encourages embracing failure as a learning opportunity and minimizing the cost of failure by testing ideas quickly and inexpensively.

For example, Google's "20% time" policy allows employees to spend a portion of their workweek on side projects, encouraging innovation and experimentation without the fear of failure.

Minimum Viable Product

- ✓ A **minimum viable product** is that version of a **new product** that allows a team to collect the maximum amount of **validated** learning about **customers** with the least effort.
- ✓ Through a cycle of **build, measure, learn**; that is the foundation of the lean **startup methodology**.

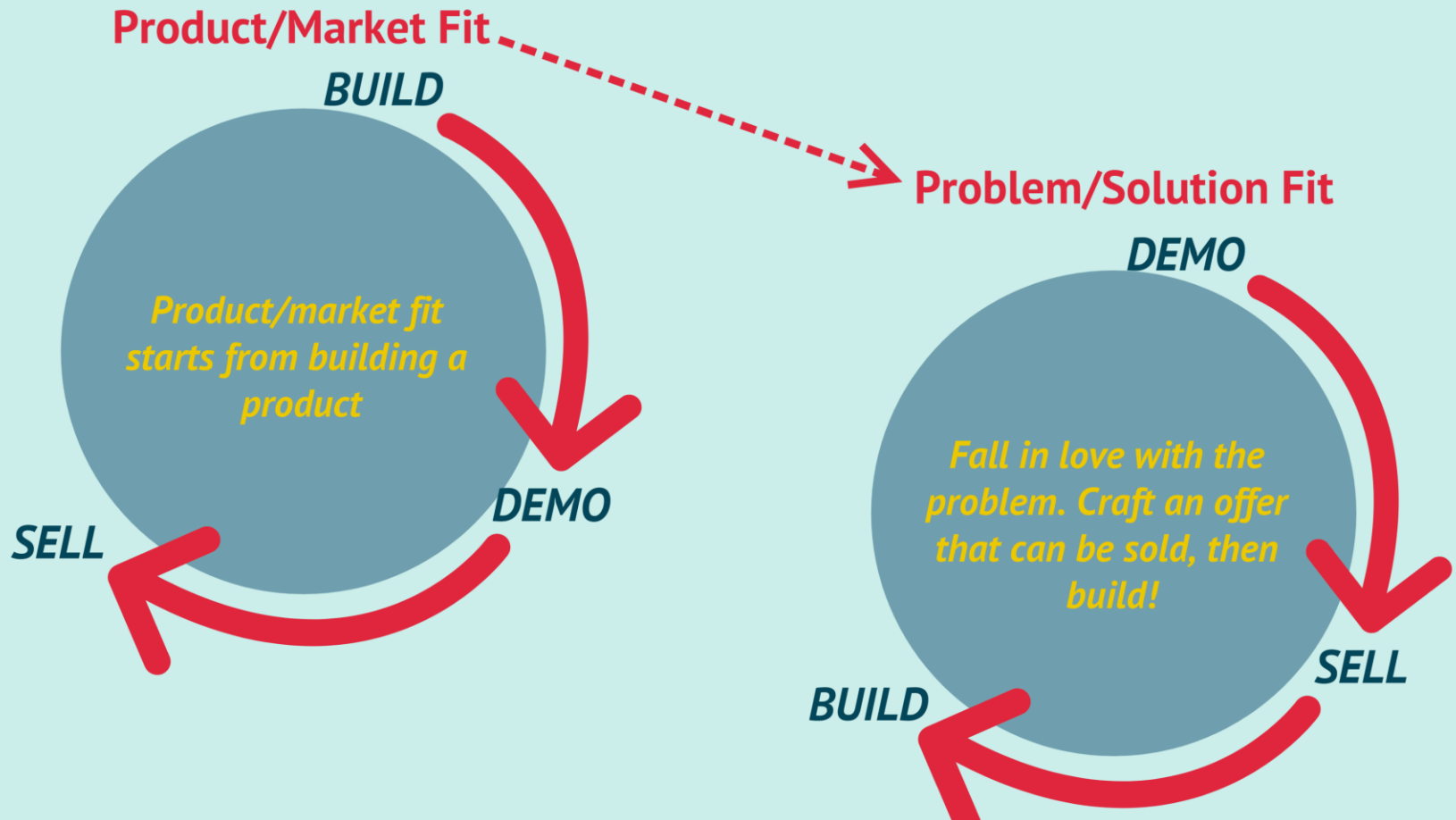
Product / Market Fit



Problem / Solution Fit

From Product/Market Fit to Problem/Solution Fit

The innovator's bias make entrepreneurs fall in the trap of getting in love with the solution rather than the problem. Instead, as pointed out by Ash Maurya, author of Running Lean, it should all start from getting in love with the problem. Thus, instead of starting from building a product, an entrepreneur might start from defining the problem, demoing a solution and selling it before the product gets built. This is the primary difference between product/market fit and problem/solution fit.



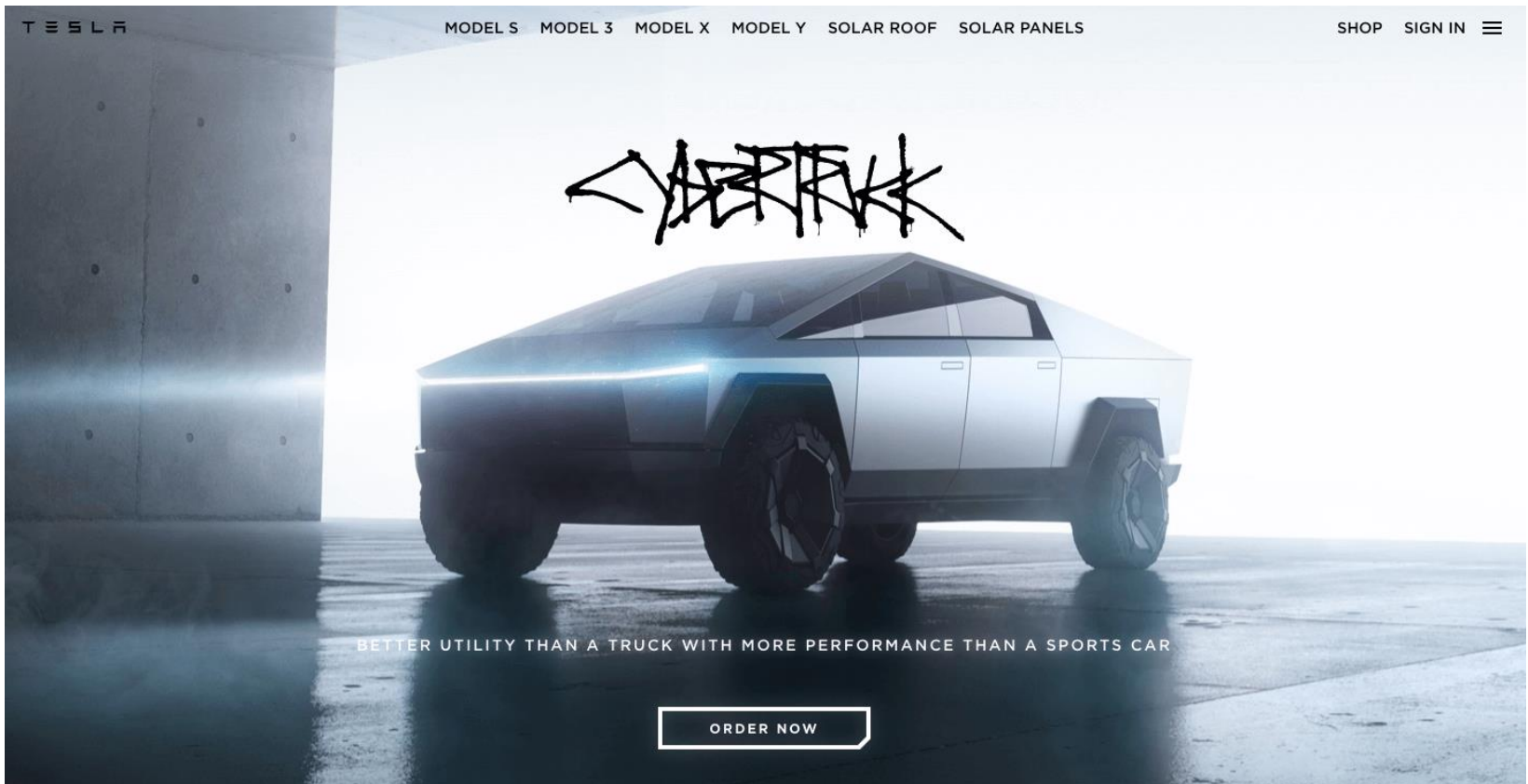
Market First

In a product-market fit scenario, we build a product first, and we iterated times and times again to find this magic moment, called “market fit.”

In MVP approach, we validate the **market first**, then build a product. So **Build** a **demo** first, **sell** that **demo**, and if you can sell the demo, then don't even build the product.

Market First

A great example of this approach is how **Tesla** pre-sells its cars by demoing them first before going into large-scale production:



Minimum Viable Product

Company	MVP Concept	Validation	Iteration
Airbnb	Online platform to rent out air mattresses and provide breakfast	Launched a simple website for hosts to list properties; the founders themselves validated it by renting out their apartment	Enhanced user experience, added secure payment options, and expanded property listings
Facebook	A social networking site for Harvard students initially	Started as a closed platform for Harvard students; expanded to other Ivy League universities	Opened up to more universities, introduced News Feed, added photo sharing, and went global
Twitter	A microblogging platform for sharing short text updates	Tested the idea internally at Odeo; initially used by employees	Introduced @mentions, hashtags, retweets, and multimedia sharing
Dropbox	File-syncing service for storing and sharing files	Created a basic video demo and shared online; received positive feedback	Enhanced features like file sharing, collaboration, and mobile access
Slack	Team communication platform with real-time messaging	Developed as an internal tool for Tiny Speck; found success and pivoted	Added integrations with other apps and expanded user base

Minimum Viable Product

Company	MVP Concept	Validation	Iteration
Amazon	Online bookstore for book enthusiasts	Launched as an online bookstore from Jeff Bezos's garage	Expanded product range, introduced Amazon Prime, and ventured into cloud computing with AWS
YouTube	Video-sharing website for sharing video clips	Uploaded the first video ("Me at the zoo") to test the platform	Improved video player, added features like comments and channels
WhatsApp	Messaging app for sending text messages over the internet	Launched to a small group of friends to test functionality	Added voice calling, video calling, and end-to-end encryption
LinkedIn	Professional networking platform for building professional connections	Launched to a small group of colleagues and friends	Expanded features to include job postings, groups, and company pages
Instagram	Photo-sharing app with filters	Released a simple iOS app that allowed users to share photos with filters	Iteratively added features like Explore, Stories, and IGTV
Uber	Ride-sharing service connecting drivers with passengers	Started with a simple app for requesting black cars; tested with friends and family	Expanded services to include UberX, UberPool, and UberEats
Spotify	Music streaming service with access to a vast music library	Launched in Sweden with a limited catalog; gathered feedback	Expanded the music library, added personalized playlists, and went international